Advanced Cube Roots Using Prime Factorization - Worksheet 2

cbr=cube root
sqr=square root

1. Find ■970299 using prime factorization. 27sqr1331

2. Find ■1889568 using prime factorization. 4sqr324

3. Find ■2985984 using prime factorization. 8sqr22703

4. Find ■7529536 using prime factorization. 8sqr117649

5. Find ■113379904 using prime factorization. 8sqr1171561

6. Find ■13481272 using prime factorization. 2sqr3370381

7. A cube-shaped warehouse has a volume of 7,529,536 m^3. Find the edge length using prime factorization. cbr235298

8. The volume of a metal cube is 13,481,272 cm^3. Determine its side length using prime factorization. cbr1685159

9. Find the cube root of $2 \times 3^3 \times 5$■ $\times 7^3$ and express the answer in prime factor form. 21cbr50

10. Find the smallest number by which 1,889,568 must be divided so that the quotient becomes a perfect cube. Then find its cube root.